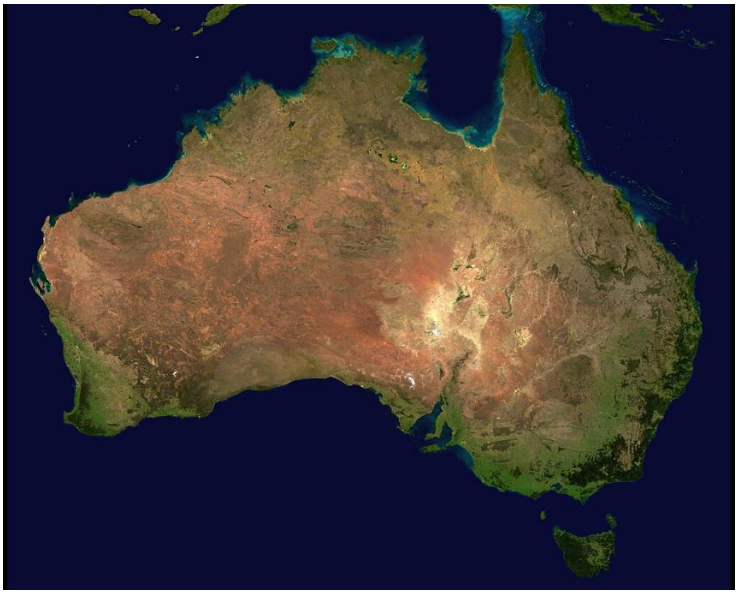
Incidence Rate of Lower Limb Amputation in Australia



STAT 495 Project

Incidence Rate Ratio Analysis and Poisson Regression Model

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1. Article Introduction

The article “Geographic Variation of the Incidence Rate of Lower Limb Amputation in Australia from 2007-12” (Dillon et al) was published on January 24, 2017. The article had the intent of understanding which factors influenced the geographic variation in the Incidence Rate of Lower Limb Amputations in Australia and developed a series of Generalized Linear Models (GLM) assuming a Poisson distribution to achieve that goal. Ethics approval for the project was granted by the South Australian Health Human Research Ethics Committee.

2. Background

What is lower limb amputation and why does it matter? The most common causes worldwide leading to lower limb amputation are diabetes mellitus, peripheral vascular disease, neuropathy, and trauma. Diabetes tends to account for just over half of these cases, with trauma hovering around 10%. Lower Limb amputation is an intensive process, and the operation to safely perform a lower extremity amputation is an exhaustive procedure requiring a full four-person medical team. Quality of life post-surgery differs wildly between different cases, but often includes chronic pain, phantom pain from the missing limb, reduced mobility, and learning to use an expensive prosthesis. The article noted that studies in other countries had shown considerable variation geographically in the Incidence Rate of Lower Limb amputation, and so hoped to quantify common risk factors for LLA (Lower Limb Amputation) and look at how those risk factors varied across regions in Australia.

3. Data Description

A total of 35,306 LLA were performed in Australia between July 2007 and June 2012, over a period of 109,042,037 person-years. Each observation represented an episode of care that also included information about geographic region, year of amputation, age group, sex, the presence of type 1 and type 2 diabetes as well as amputation level. A limitation of this type of data was that it only described LLA episodes, so there was no ability to delineate between first and subsequent procedures. This “memory-less” property meant it was impossible to determine repeat offenders, and analyzing the data from a raw poisson incidence rate ignored a high potential for repeat offenders (for example, a patient with severe Type 2 diabetes could be at risk for several individual toe amputations in sequential incidents). In addition, several potential risk factors were removed from the input variable pool to increase statistical power so less populated regions could be included. Across the time series of this investigation, the crude IR-LLA was 32.4 per 100,000 population (per year).

An analysis of the raw data summary leads to the following notable observations. Toe amputations accounted for 40% of all LLA, and three-quarters were below ankle procedures. Half of all LLA patients had Type 2 Diabetes Mellitus (compared to about 5% of the general population having Type 2 Diabetes. Most notably, two-thirds of all LLA were performed on

males. In addition, two thirds of LLA were performed on people over the age of 60 and nearly none below the age of 35. This implies that our age variable coefficient (to be discussed later) loses statistical power once talking about younger adults below the age of 35.

4. Results

Since this is an analysis of an article also, I want to make sure to differentiate between conclusions made by my own statistical inference and statistical conclusions made by the article. The following two paragraphs are from my own conclusions. We are 99% sure the true population incidence rate for Lower Limb Amputation in Australia is between 31.9362 and 32.8248 per 100,000 population years. The estimated incidence rate of Lower Limb Amputation for females in Australia is 47.534% of the expected rate for males in Australia. Our 99% CI for Relative Risk LLA from females to males is (.46162,.48946), which means we can convincingly reject a null hypothesis that both genders suffered LLA equally, and instead state males suffer from LLA at a much higher rate than females (at least 2 to 1). Our own Poisson Regression for Incidence Rate per 100,000 person years for Lower Limb Amputations in the territory of New South Wales is

λ^0= exp(3.027017 + .00995 * age + .19062 * male + .10436 * Type2 + .00995 * Type1 - .04082 * (08/09) - .06187 (09/10) - .07257 (10/11) - .08338 (11/12))

Where age is the patients age in years, male is 1 if the patient is male and 0 if the patient is female, Type2 and Type1 are 1 if the patient has the respective type of diabetes and 0 if not, and the year periods are binary variables with the first data collection year (07/08) being the baseline. Patient's age, sex, and the indicator of Type 2 Diabetes are significant predictors of the incidence rate of Lower Limb Amputations at the 5% significance level. For a one-year increase in the patient's age, the estimated average number of expected LLA increases by (exp(.00995) - 1) * 100% = .99%. The estimated average number of LLA for patients who have Type 2 Diabetes is exp(.10426) * 100% = 111% of that for patients without it. Also, the estimated average number of LLA for patients who are male is exp(.19062) * 100%= 120.99% of that for female patients.

If we want to predict a value for expected lower limb amputations rate for hypothetical Harold, a 60-year-old man living in Sydney (New South Wales from 2008 to 2009, with Type 2 but not Type 1 diabetes, we plug his information into the IR model.

λ^0= exp(3.027017 + .00995 * 60 + .19062 * 1 + .10436 * Type2 -.04082 * 1) = 48.336

We get a result of a predicted 48.336 LLA per 100,000 person years. Another way of interpreting this value is we expect Harold to have a .04833% chance of undergoing a LLA during his year in Sydney, about a 1 in 2,000 chance.

The following statistical interactions are taken from the article, with my own commentary and analysis added on. There was notably a significant interaction between sex and type 2 diabetes. The presence of type 2 diabetes increased the RR more in females (RR 1.73; 95%CI 1.55–1.93)

than in males (RR 1.32; 95%CI 1.04–1.67). Relative Risk did NOT change significantly with respect to time (suggesting problem not improving). While the point estimate for Incidence Rate was slowly going down by year, it was not enough for a statistical conclusion. The Northern Territory region of Australia had by far the highest IR for LLA (statistically significant). However, the interaction between gender was REVERSED in the Northern Territory: in the Northern Territory being male reduced the RR by 10% (RR 0.90, 95%CI 0.83–0.98).

5. Conclusion

Aside from the less populated and divergent Northern Territory, the article and findings imply that Incidence Rates for Lower Limb Amputations DO NOT differ much across different states in Australia. The report notes "we suggest that part of the variation in the IR-LLA between regions could be explained by the differing proportions of people with diabetes in those regions (r^2 = 0.44; $F_{1,25}$ = 19.84, $p<0.001$)." (Discussion: Paragraph 2)

If I oversaw research funding in Australia, I would promote a follow up research project that would specifically send statisticians to the Northern Territory to gleam more knowledge from these noteworthy interactions. In addition, because the presence of Type 2 Diabetes and Gender were by far the two most significant predictors of Lower Limb Amputations, I would increase funding for awareness campaigns on prevention methods for LLA to be placed in diabetes clinics and men's health resource locations across the country.

There were many limitations on the type of data access that impaired the predictive power of the study. Because of ethics board limitations resulting in anonymous data, the memoryless quality of data leads to a failure with the study to identify and categorize repeat offenders. Also, the lack of predictive power within the patient-age variable and omission of several input variables that could have had key underlying predictive power, all lead me to believe there is a more complicated underlying model that could further identify the roots of Lower Limb Amputations.

Appendices

a) SAS Code

```
data cases;
input T n;
cards;
109042037 35306
;

%let conf_level=99;
data CI (drop=alpha);
set cases;
alpha=(1-0.01*&conf_level);
lambda_hat=n/T;
lambdaL=gaminv(alpha/2,n)/T;
lambdaU=gaminv(1-alpha/2,n+1)/T;
run;
proc print data=CI noobs;
run;
```

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```
data cases;
input n1 T1 n2 T2;
cards;
11453 54795459 23853 54246578
;
%let conf_level=99;
data IRR_CI;
set cases;
IRR=(n1/T1)/(n2/T2);
z=-probit((1-0.01*&conf_level)/2)
SE=sqrt(1/n1+1/n2);
LCL=IRR/exp(z*SE);
UCL=IRR*exp(z*SE);
keep IRR LCL UCL;
run;
proc print data=IRR_CI noobs;
run;
```

b) SAS Results

The SAS System

lambda_hat	lambdaL	lambdaU
.000323783	.000319362	.000328248

The SAS System

IRR	LCL	UCL
0.47534	0.46162	0.48946

c) R Code

```
#exact Poisson confidence interval
install.packages("DescTools")
library(DescTools)
PoissonCI(35306, 109042037, method="exact", conf.level = 0.99)
```

```
n1<- 11453
T1<- 54795459
n2<- 23853
T2<- 54246578
conf.level<- 99
z<- -qnorm((1-0.01*conf.level)/2)
SE<- sqrt(1/n1+1/n2)
print(IRR<- (n1/T1)/(n2/T2))
print(LCL<- IRR/exp(z*SE))
print(UCL<- IRR*exp(z*SE))
```

d) R Results

```
> PoissonCI(35306, 109042037, method="exact", conf.level = 0.99)
              est      lwr.ci       upr.ci
[1,] 0.0003237834 0.000319362 0.0003282485
```

```
> print(IRR<- (n1/T1)/(n2/T2))
[1] 0.4753396
> print(LCL<- IRR/exp(z*SE))
[1] 0.4616223
> print(UCL<- IRR*exp(z*SE))
[1] 0.4894646
>
```

References

Geographic Variation of the Incidence Rate of Lower Limb Amputation in Australia from 2007-12 (Dillon et al) https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0170705